Fundamental CS for AI Era

Building human-centered digital products in the AI era

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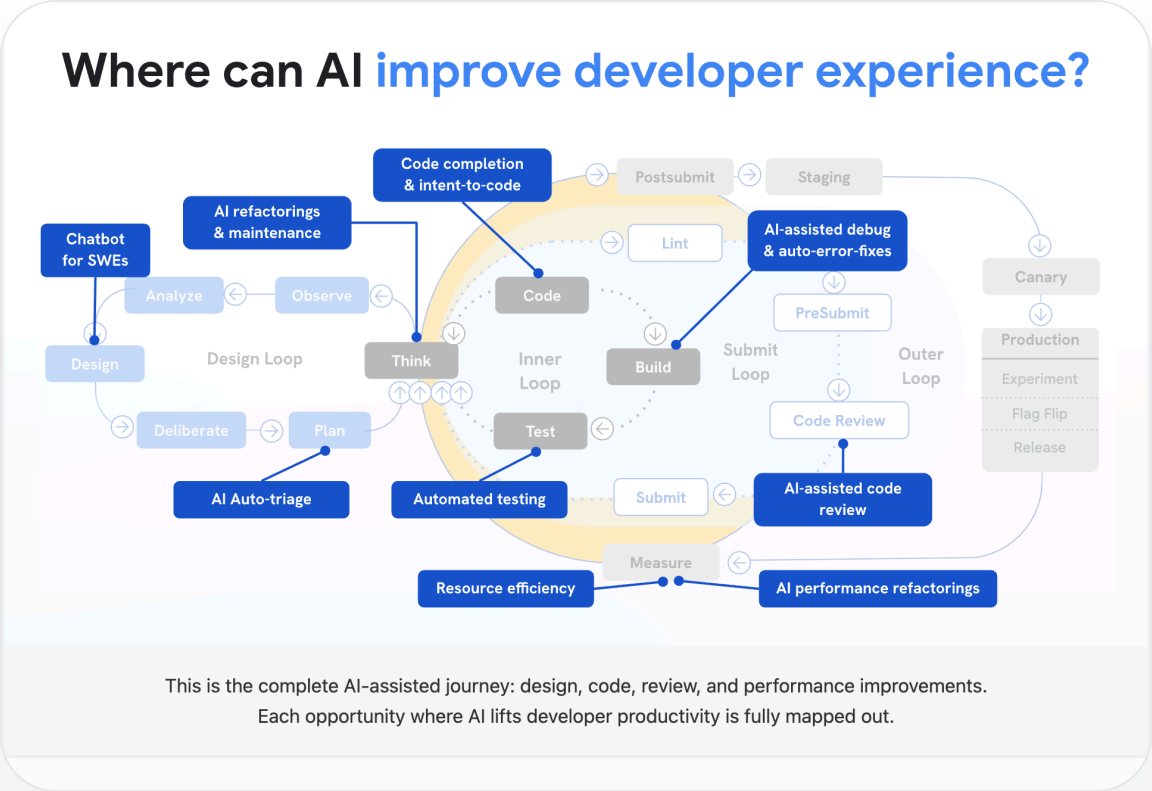
AI makes code generation fast and accessible

Anyone in this room can ship working code today — developer or not.

Generating code is no longer the hard part. **We all agree on that.**



P3 · COMING PROBLEM

Understanding becomes the bottleneck

AI generates code faster than we can understand it.

The constraint moves from **writing** code to **understanding what changed, why it works, and what it might break.**

Geoffrey Litt · 2 July 2026

> **Hot take: I think it's still important to understand the code that our agents write!**

Understanding matters not just to verify, but to participate. — "Understanding is the new bottleneck"

Imagine calm confidence in any codebase

Opening an unfamiliar repository without dread.

Reviewing an AI-generated PR and **knowing** what to look at first.

That feeling is trainable — and AI itself can train it.

AI × CS

The win is not “skip fundamentals.” The win is “learn fundamentals faster, with feedback.”

P5 · FALSE HOPE

“AI means I can skip fundamentals”

Tempting — but this is what it looks like from the receiving end:

Dex Horthy · 20 June 2026

> **If people are tokenmaxxing bugs into production with kLOC PRs that they didn’t read themselves, those people shouldn’t have jobs.**

“Coaching juniors on SWE fundamentals is hard work and it takes time.”

P6 · AUDACIOUS REALITY

Fundamentals are the AI multiplier

Data structures, algorithms, systems thinking, debugging, trade-offs.

They give you the **taste** to judge AI output and the **words** to steer it.

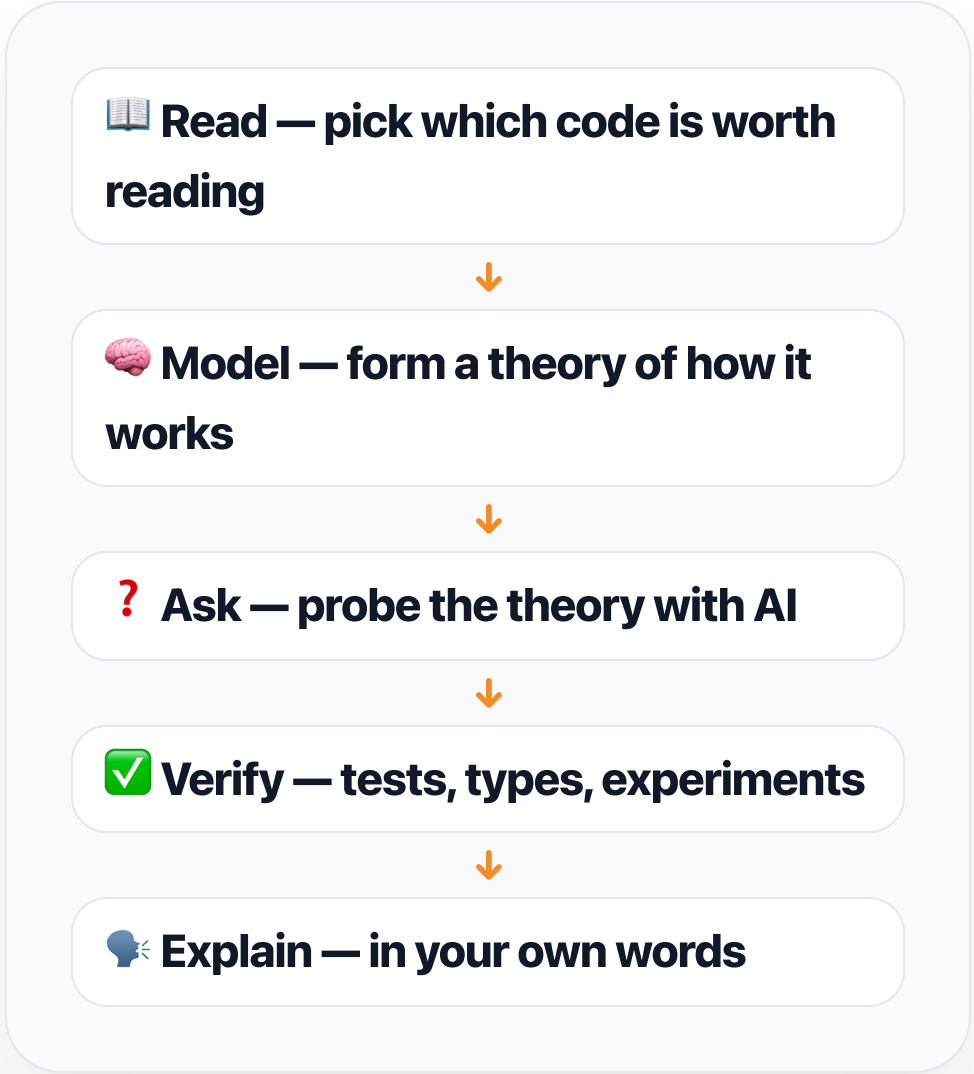
Gergely Orosz · 2 July 2026

> **If you don't know what good code looks like, you will have no idea if what the models generate are any good.**

"This is exactly why experienced software engineers are valuable and will be valuable." (via @mitchellh)

A loop anyone can run

Not a talent. A habit — five moves, repeated on every change:



Your first doable step: `/teach`

An open-source Claude Code skill that turns that loop into a **persistent learning workspace**.

1. Copy `skills/teach` from **github.com/zainfathoni/agent-workflows** into `~/.claude/skills`
2. Run `/teach` with any topic you want to learn
3. Answer the mission interview, take one lesson and its quiz
4. Come back tomorrow — it remembers where you left off

🎯 **MISSION.md — why you're learning**

📚 **lessons/ — short, single-win lessons**

📝 **quizzes — speed regulators for real understanding**

🗂️ **learning-records/ — evidenced progress**

Quizzes as speed regulators — the same technique Geoffrey Litt proposes for understanding AI code.

You can learn anything you want

Not just CS fundamentals. The same workspace teaches you:

- a new framework or language
- an unfamiliar codebase, one PR at a time
- algorithms and system design
- even non-code topics

Instead of:

passive tutorials you abandon by chapter three.

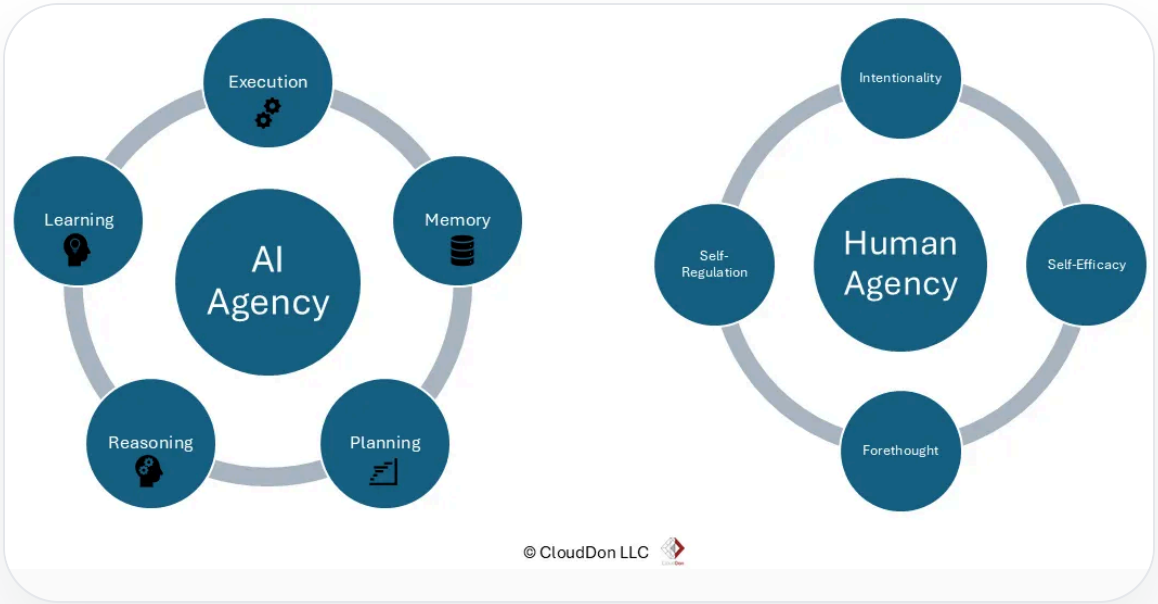
You get:

a personal curriculum with lessons, quizzes, and memory — paced by what you actually retain.

Engineers become better problem framers

In the AI era, human-centered builders are not the people who type the most code.

They are the people who can **frame problems clearly enough for humans and machines to solve together.**



Live Demo

Let's learn something together — for real, for the next hour.

🙋 You pick the topic. You answer the quizzes.

How the next hour works

1. **Pick** (10 min) — you shout topics, we vote on one
2. **Mission** (15 min) — `/teach` interviews us; you answer, I type
3. **Learn** (20 min) — we take the first generated lesson together
4. **Quiz** (15 min) — you answer; wrong answers steer the next lesson

Ground rule:

I will not edit the AI's questions or lessons. What you see is what the skill does.

Fallback topic:

“How does the code AI just wrote for me actually work?” — the theme of this talk.

What we'll watch it build

A learning workspace, growing live on screen:

- `MISSION.md` — the room's shared goal
- `RESOURCES.md` — sources we trust
- `lessons/*.html` — one win per lesson
- `learning-records/*.md` — proof of what we learned

> **Understanding matters not just to verify, but to participate.**
>
> Geoffrey Litt — this hour is that idea, practiced.

References

- Geoffrey Litt — Understanding is the new bottleneck
- Dex Horthy — against lazy engineering
- Gergely Orosz — knowing what good code looks like
- Zain Fathoni — A to Z #3: my personal experience
- The `/teach` skill — github.com/zainfathoni/agent-workflows
- Dan Roam — The Pop-up Pitch

> **Fundamentals are not the past.**
>
> **They are how we steer the future.**

Q & A

<https://zainf.dev/fundamental-computer-science-ai-era>

<https://github.com/zainfathoni/agent-workflows>

<https://zainf.dev/a-z-3>